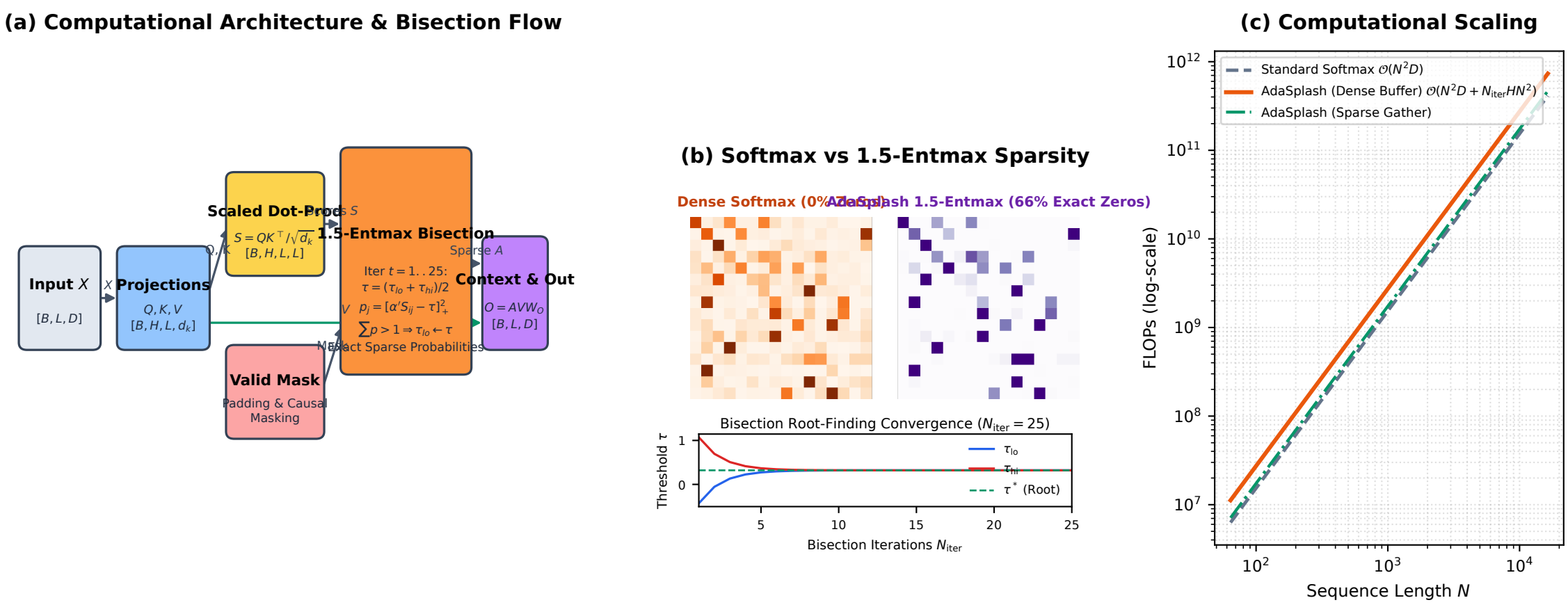

(a) Computational Architecture & Bisection Flow
Input X
$[B, L, D]$
X
Projections
Q, K, V
$[B, H, L, d_k]$
Q, K
Scaled Dot-Prod
$S = QK^\top/\sqrt{d_k}$
$[B, H, L, L]$
Scores S
1.5-Entmax Bisection
Iter $t = 1..25$:
$\tau = (\tau_{lo} + \tau_{hi})/2$
$p_j = [\alpha' S_{ij} - \tau]_+^2$
$\sum p > 1 \Rightarrow \tau_{lo} \leftarrow \tau$
Exact Sparse Probabilities
Sparse A
V
Context & Out
$O = AVW_O$
$[B, L, D]$
Mask
Valid Mask
Padding & Causal Masking
(b) Softmax vs 1.5-Entmax Sparsity
Dense Softmax (0% Sparsity)
AdaSplash 1.5-Entmax (66% Exact Zeros)
Bisection Root-Finding Convergence ($N_{\text{iter}} = 25$)
Threshold τ
1
0
5
10
15
20
25
Bisection Iterations N_{iter}
τ_{lo}
τ_{hi}
τ^* (Root)
(c) Computational Scaling
Standard Softmax $\mathcal{O}(N^2 D)$
AdaSplash (Dense Buffer) $\mathcal{O}(N^2 D + N_{\text{iter}} H N^2)$
AdaSplash (Sparse Gather)
FLOPs (log-scale)
10^{12}
10^{11}
10^{10}
10^{9}
10^{8}
10^{7}
10^{2}
10^{3}
10^{4}
Sequence Length N